

HACKOUT “26

Ideation

Waste-to-Carbon-Value Chain Tracker

What is Our Problem??

Most of the wastes produced by human practices today such as farming, manufacturing etc are dumped in landfills, where they decay and release methane which in-turn increase the concentration of Greenhouse gases causing global warming.

Meanwhile the very plants made to utilize and recycle organic wastes, such as biochar kilns and biogas digestors are not used to their capacity.

This is a clear waste of resources and infrastructure that could be used for betterment of humanity.

So we decided to work upon it.

Why does India need it the most??

India currently ranks just behind China in number 2 for being one of the most wasteful countries in matter of Organic Waste which is very useful conversion into biochar, biogas, or carbon-negative materials.

This is very crucial as our work can contribute a lot in this is problem.

According to Central Pollution Control Board (CPCB) Annual Reports on Solid Waste Management Rules (SWM Rules, 2016), which estimate India's urban MSW generation at ~1.5 to 1.7 lakh tonnes per day (TPD).

Approximately 50-55% waste contains high moisture and low calorific value which is best for conversion into biochar, biogas, or carbon-negative materials.

Why would people adopt this??

But when u look at this issue from the side of farmer or someone who barely makes ends meet the bigger picture often looks blurry. It wouldn't make sense for a farmer to go out of his way and send his organic waste for further processing.

So our goal is to make sure we remove this friction as much as possible.

1. Provide compensation to entities interested in our project.
2. Implementation of Simple application allowing users to have less effort learning our system.

Problems Faced in Implementation

1. Drivers Skipping Pickups or Taking Long Detours

The Issue: Contracted waste drivers skip scheduled pickups or take inefficient routes, burning unnecessary fuel and leaving perishable waste stranded.

The Solution: Automated Geofence Verification. The app automatically logs arrival and departure timestamps only when the driver's phone/vehicle GPS enters a 50-meter geofence of the pickup location, releasing payouts only for verified visits.

2. Dirty Bins and Foul Odor at Collection Sites

The Issue: Wet waste bins begin smelling within hours, attracting pests and making hotel kitchens or housing societies abandon the collection system.

The Solution: Bin-Swap System. The logistics team picks up the full bin and immediately replaces it with a pre-sanitized, empty bin rather than emptying and washing it on-site.

3. Mixed Waste in Segregated Bins

The Issue: Staff throw plastic bottles or cutlery into organic bins out of convenience, ruining the batch.

The Solution: Photo Check Before Pickup. The generator must upload a quick overhead photo of the open bin in the app before booking a pickup; basic image scanning or driver verification flags non-segregated loads before the truck is dispatched.

4. Weight Disagreements Between Sender and Receiver

The Issue: The waste generator claims they handed over 100 kg, but the processing plant logs only 80 kg, leading to disputes over fees and carbon credits.

The Solution: Digital Weighbridge Handshake. Drivers carry a portable hanging hook scale for small loads or pass through an automated weighbridge at the facility gate, generating a single tamper-proof receipt sent instantly via SMS to both parties.

5. Staff Forgetting Daily Pickup Requests

The Issue: Canteen or market workers forget to open the app and schedule waste dispatch during busy rush hours.

The Solution: Automated WhatsApp Reminders. A scheduled bot pings the facility manager at a fixed daily hour (e.g., 9:00 PM) asking for a quick reply (e.g., "Reply 1 for Full Bin, 2 for Half Bin"), automatically queueing next morning's route.

Revenue Model

1. Waste Marketplace Cut (5%–8%)

Take a direct commission on every tonne of waste or biomass bought by biochar or biogas plants through the app.

2. Delivery Fee Margin (10%–15%)

Batch nearby pickups and combine return trips, keeping a percentage of the saved fuel and hauling costs.

3. Monthly Compliance Fee (₹3,000–₹10,000/month)

Charge hotels, markets, and factories a software subscription for official waste receipts that protect them from municipal fines.

4. Carbon Credit Cut (15%–20%)

Calculate and verify the climate savings (avoided landfill methane and stored biochar) and take a cut when carbon credits are sold.

Example per 100 Tonnes Managed

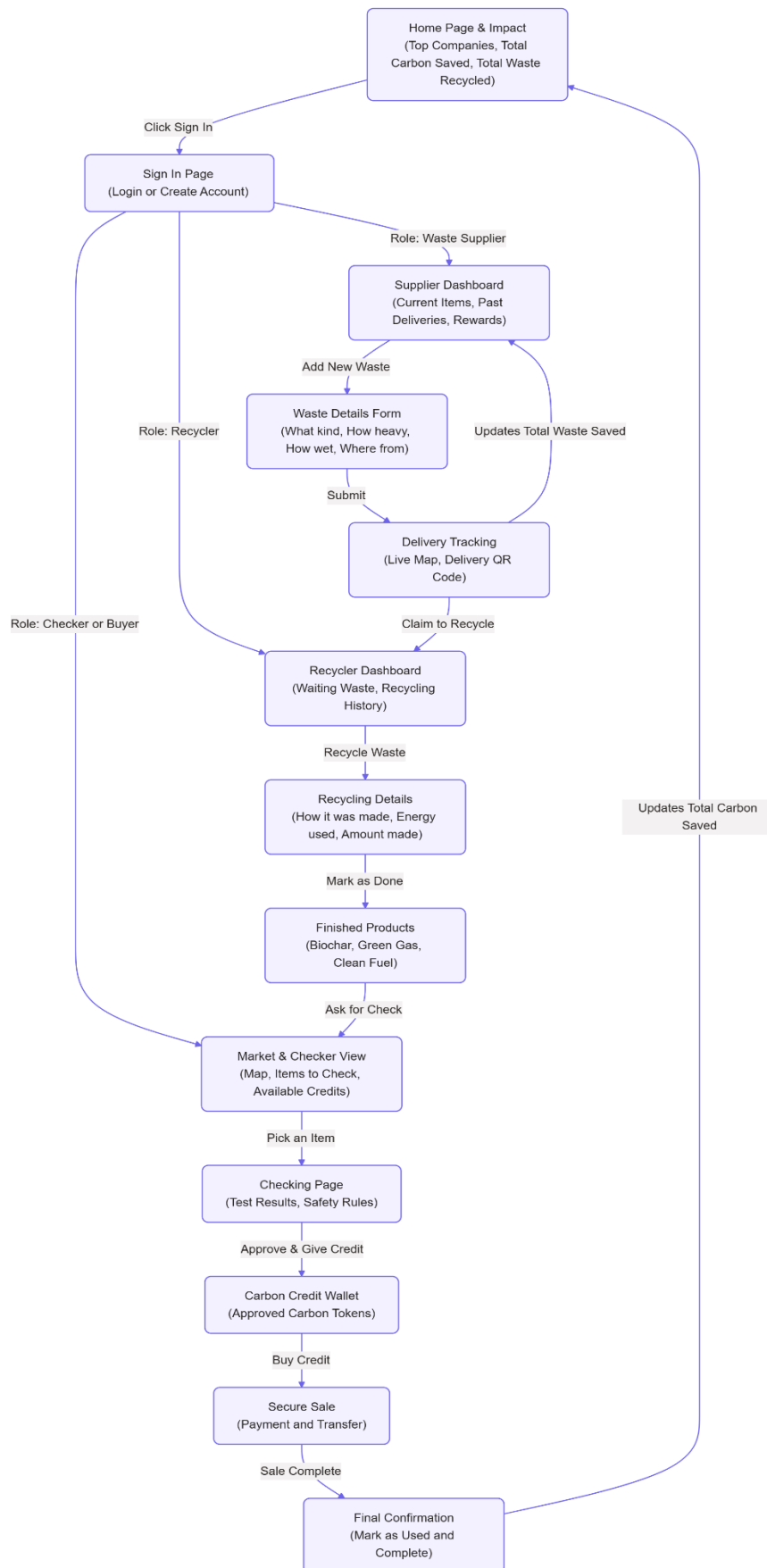
Trading cut: ₹7,000

Logistics margin: ₹4,500

Carbon credit share: ₹15,000

Total made: ~₹26,500 per 100 tonnes diverted.

Application Model



Waste-to-Carbon Tracking Platform Summary

This platform provides an end-to-end digital ledger that traces raw waste from its point of origin through its physical transformation into certified carbon credits. By connecting all parties in the supply chain, it ensures transparency, accountability, and verifiable environmental impact.

1. Waste Generation & Logistics (Suppliers)

Waste suppliers initiate the cycle by logging raw materials into the system, detailing specific metrics like waste type, weight, and moisture content. The platform generates a tracking code and monitors the shipment to a recycling facility, instantly updating the public dashboard's total waste diverted.

2. Transformation & Inventory (Recyclers)

Upon claiming and receiving the waste, recycling facilities process the materials. They log the exact conversion details into the system, including the recycling method used, energy consumed, and the final yield of physical sustainable products (such as biochar, green gas, or clean fuel).

3. Audit & Certification (Checkers)

Before any environmental claims are validated, independent checkers review the facility's production logs, lab test results, and safety compliance. Once the data is thoroughly verified and approved, the system mints official, digital Carbon Credits and deposits them into a secure wallet.

4. Secure Trading & Impact (Buyers)

Buyers purchase these verified credits on the platform's open marketplace via a secure transaction. Once a credit is bought and officially "retired" (meaning it cannot be resold), the public leaderboard is updated in real-time to reflect the total carbon emissions successfully offset.

Result: A fully transparent ledger that updates public impact metrics (Total Carbon Saved, Total Waste Recycled) in real-time.

Tech Stack

Frontend Layer

Core Framework: React + TypeScript (Vite or Next.js App Router).

Styling & UI: Tailwind CSS + shadcn/ui

Backend & Data Layer Database:

Supabase PostgreSQL with the PostGIS extension enabled

Authentication: Supabase Auth.

Storage: Supabase Storage bucket for storing weighbridge receipts and pre-dispatch waste verification photos.